Predicting vortex shedding behind a circular cylinder with physics-informed neural networks

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Abstract

A data-free physics-informed neural network (PINN) applied to the unsteady wake of a circular cylinder is known to collapse onto the steady symmetric solution: the steady field satisfies the governing equations exactly, and the training loss has no notion of its physical instability. This project reproduces that collapse on purpose-generated, validated ground truth, and then determines which of three candidate mechanisms defeats it. On a lattice Boltzmann dataset at Re = 100 (C_d within 0.3% and St within 2.3% of published benchmarks), a plain PINN retains 10.6% of the reference wake amplitude. Adding Fourier features or a Reynolds decomposition does not help. *Causal time-marching alone restores shedding data-free*, at 89.2% retained amplitude, and an equal-budget control rules out extra optimiser steps as the cause. The result holds at every Reynolds number tested from 60 to 180: the control keeps 10–43% of the wake, the mechanism-equipped network 88–104%. The best data-free configuration, marched over five shedding periods, predicts the shedding frequency to within 0.1%, at 94.8% of the correct amplitude and 2.4° of phase, with a velocity-field error of 3.6%. A second contribution adds a pressure-free physics residual to a Fourier Neural Operator trained after the method of Renn et al.; the physics term lowers forecast error by 20–28% out to one shedding period, after which both variants have failed.

1 Introduction and objective

Above Re $\approx$ 47 the wake behind a circular cylinder becomes unstable and sheds a Kármán vortex street at a frequency characterised by the Strouhal number St $= fD/U$. Predicting that shedding with a neural network is not merely a question of accuracy, because of a structural difficulty:

1. **The steady solution remains a valid solution.** A symmetric, non-shedding field satisfies the Navier–Stokes equations and every boundary condition exactly. It is unstable physically, but a PINN loss measures residuals, not stability, so the steady field is an easily reachable minimum.

2. **Spectral bias.** Networks fit low-frequency content long before high-frequency content, which directly opposes representing an oscillatory wake.

3. **The initial condition is a measure-zero slice** of the space–time domain, and is therefore weakly weighted against the bulk PDE residual that fills it.

The documented consequence [2] is that a data-free PINN "behaves like a steady solver", while a data-driven one sheds only where data is present. The objective of this project is to reproduce that failure on validated ground truth of our own, and then to establish *which* mechanism defeats it, with an ablation controlled well enough that the answer is attributable.

A second thread connects the work to its reference paper. Renn et al. [1] forecast subcritical cylinder wakes with a Fourier Neural Operator (FNO) that is *purely data-driven* — there is no physics term anywhere in it. Whether adding physics to such an operator improves forecast stability is the natural question joining the two topics, and is addressed in Section 8.

Reynolds number. All results are at $\mathrm{Re} = 100$: laminar and unambiguously two-dimensional. The reference paper's $\mathrm{Re} = 240$–3060 is genuinely three-dimensional, where a 2-D model would be solving the wrong physics. Section 7 repeats the central comparison across $\mathrm{Re} = 60$–180.

Software. The work builds on `underPINN`, a JAX PINN framework, which supplies the network architectures, the geometry primitives and a steady Navier–Stokes residual. The unsteady operator (Section 3), the lattice Boltzmann solver, the ablation machinery and the operator-forecasting code are contributions of this project.

2 Ground truth by lattice Boltzmann

No experimental data was available, so the reference wake is generated. The solver (`src/cfd/lbm_cylinder.py`) is a D2Q9 two-relaxation-time (TRT) lattice Boltzmann method written in JAX.

Why these particular choices.

- **TRT collision.** A single-relaxation-time (BGK) scheme places the bounce-back wall at a viscosity-dependent position, so the cylinder's effective diameter drifts with Re. TRT relaxes the symmetric and antisymmetric parts separately, allowing the magic parameter $\Lambda = (\tau^+ - 1/2)(\tau^- - 1/2)$ to be pinned at $3/16$, which fixes the wall location.
- **Half-way bounce-back** for the no-slip cylinder, placing the wall midway between nodes.
- **Momentum-exchange** for the surface force, which is what makes C_d accurate enough to validate against published values.
- **Specular (free-slip) top and bottom walls**, enforcing $v = 0$ exactly and matching the symmetry condition used on the PINN side.

Shedding is triggered by a brief transverse inlet perturbation; without it the symmetric base flow persists indefinitely — the same degeneracy that the PINN later falls into, appearing here in the reference solver.

Production dataset. $\mathrm{Re} = 100$, $D = 24$ lattice units, $\mathrm{Ma} = 0.087$, grid 720×1200, blockage $D/H = 2\%$. The saved record is 641 snapshots spanning $t^* = 91$–212 on a 241×121 grid at $\Delta x = 0.083D$ and $\Delta t^* = 0.19$, chosen near the reference paper's PIV resolution so that the later comparison is like-for-like.

Table 1: Validation of the ground truth at $\mathrm{Re} = 100$.

quantity	computed	benchmark	error
C_d (mean drag coefficient)	1.3798	1.37–1.38	+0.3%
St (Strouhal number)	0.1611	0.165	−2.3%
divergence error	0.55% of $\lvert\nabla\mathbf{u}\rvert$, tracking $O(\mathrm{Ma}^2)$		

Convergence study. Two candidate error sources were separated (Figure 1). Varying resolution from $D = 16$ to $D = 40$ at fixed blockage moves C_d by only 2.4%: the grid is converged. Varying blockage at fixed resolution moves it far more, $C_d = 1.627$, 1.459 and 1.431 at 5.0%, 3.3% and 2.5%, extrapolating to 1.348 at zero blockage (−2.0% against the benchmark). *The apparent +18% error at 5% blockage is domain confinement, not a solver defect* — the headline validation result.

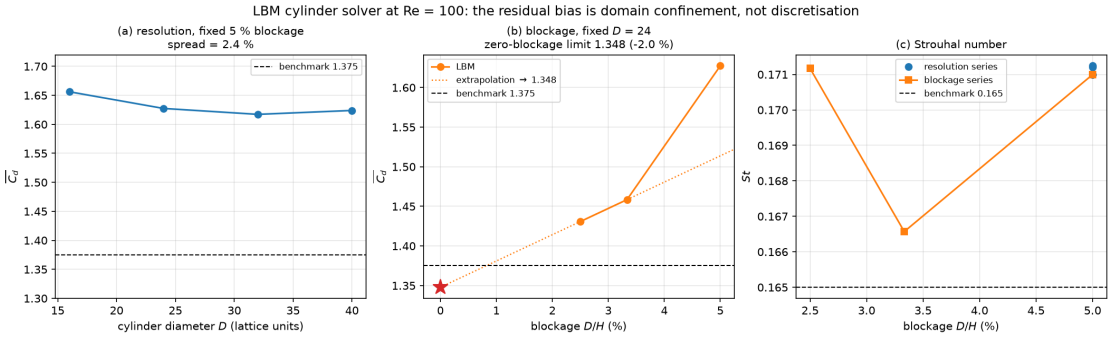


Figure 1: Solver validation: resolution series, blockage series, and the extrapolation to zero blockage.

A methodological note. An early estimate of St taken from an FFT peak gave a +0.3% agreement that was pure luck: with ∼12 recorded periods the FFT bin width (0.014) exceeds the discrepancy being measured. All St values reported here come from a zero-crossing estimator, which resolves the period to a fraction of a time step; the honest error is −2.3%.

3 An unsteady Navier–Stokes operator

`underPINN` ships only a steady incompressible residual, which cannot represent shedding at all. This project supplies the unsteady operator (`src/pinn/ns2d_unsteady.py`), conforming to the framework's `BasePDE` contract so it composes with the existing solvers, samplers and networks. Non-dimensionalised on D and U,

$$u_x + v_y = 0, \tag{1}$$

$$u_t + uu_x + vu_y + p_x - \tfrac{1}{\mathrm{Re}}(u_{xx} + u_{yy}) = 0, \tag{2}$$

$$v_t + uv_x + vv_y + p_y - \tfrac{1}{\mathrm{Re}}(v_{xx} + v_{yy}) = 0, \tag{3}$$

with the network mapping $(x, y, t) \mapsto (u, v, p)$.

Two derivative backends are implemented: a literal `jax.hessian` reference, and a forward-over-forward JVP path computing only the six derivative groups the residual needs, where a full Hessian would compute 27.

Table 2: Verification of the unsteady operator. All three checks pass at machine precision.

check	result
annihilates the Taylor–Green vortex (an exact NS solution)	1.6×10^{-16}
fast JVP backend versus literal Hessian	2.8×10^{-17}
steady limit versus `underPINN`'s own `NavierStokesPDE`	5.6×10^{-17}
speed-up of the fast backend	2.5×

The third check is the important one: it shows the new operator agrees with the framework's existing physics, rather than being a parallel implementation that merely looks correct.

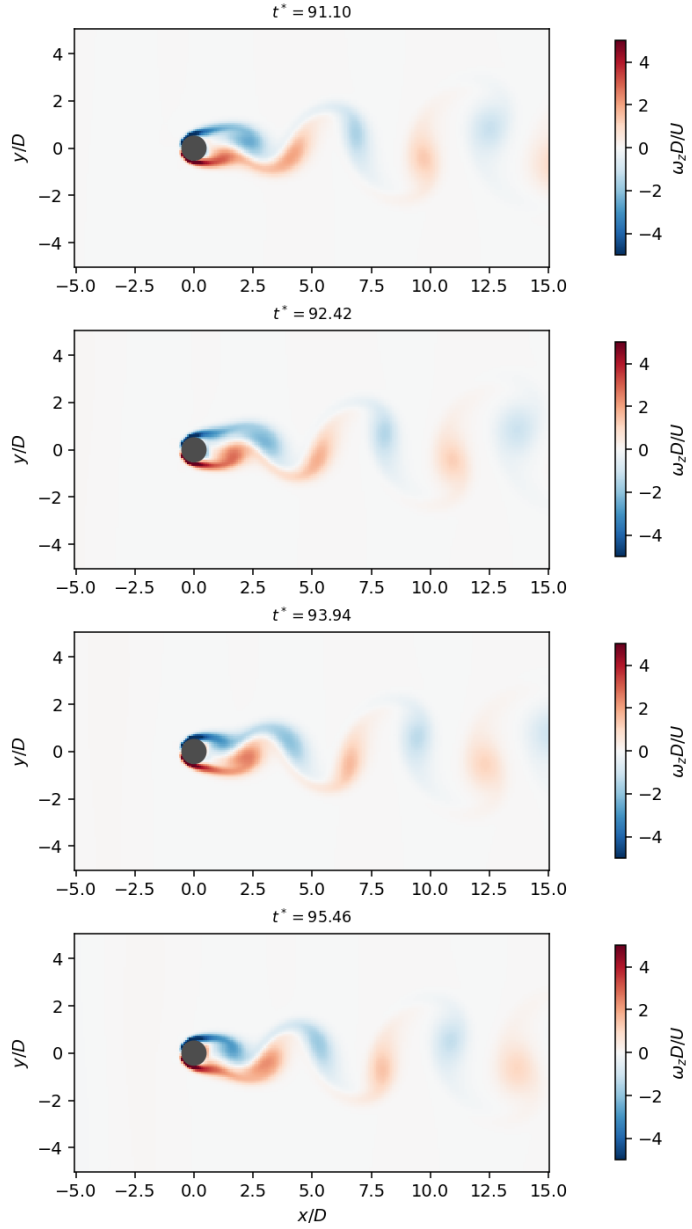


Figure 2: The reference Kármán street over one shedding period.

4 Reproducing the collapse

A plain multilayer perceptron $[3, 128^{\times 6}, 3]$ is trained on physics, boundary and initial conditions with *no data*, over a single time window of one shedding period, for 20 000 epochs.

Shedding retained is this project's own metric: the ratio of transverse-velocity r.m.s. at a fixed wake probe between the PINN and the LBM reference. Near 0% means collapse to steady flow; 100% means the wake amplitude matches.

The wake keeps about a tenth of the reference amplitude — the PINN is effectively solving the steady problem. Quadrupling the budget to 80 000 epochs moves this only to 21.9%, so the collapse is not an artefact of under-training.

Table 3: The documented failure, reproduced. Two seeds.

quantity	seed 0	seed 1
shedding retained	9.8%	11.3%
relative L_2 on (u, v)	0.184	0.183

5 Defeating the collapse

Three mechanisms are implemented as switches on one trainer, so that the ablation is controlled — every configuration shares the sampler, the optimiser, the schedule and the evaluation.

Table 4: The ablation matrix: what each configuration attacks.

configuration	mechanism	obstacle attacked
`baseline`	—	(control)
`fourier`	random Fourier features	spectral bias
`decompose`	Reynolds decomposition $u = u_0 + u'$	conditioning, dynamic range
`march`	causal time-marching	weak initial condition, causality
`all`	all three	—
`all+data`	plus 4 000 supervised points per window	—

Two implementation points matter. First, the Reynolds decomposition requires a *differentiable* mean field, because the momentum residual contains ∇u_0 and $\nabla^2 u_0$; bilinear interpolation of a grid gives piecewise-constant first derivatives and identically zero second derivatives. The mean is therefore fitted by a separate frozen network (relative $L_2 = 1.1 \times 10^{-3}$). Second, that mean is obtained by *regression on the measured mean*, not by a steady PINN solve, because the time mean of a shedding flow does not satisfy the steady equations — it carries the divergence of the Reynolds stresses.

5.1 Result

Table 5: Full-budget ablation at $\mathrm{Re} = 100$: 20 000 epochs per window, two seeds, seed means. *Time-marching alone is sufficient, data-free.*

configuration	shedding retained	relative L_2
`baseline`	10.6%	0.183
`fourier`	9.7%	0.167
`decompose`	7.0%	0.110
`march`	**89.2%**	0.078
`all`	**90.2%**	0.047
`all+data`	100.0%	**0.015**

Neither spectral bias nor conditioning was the binding constraint. Both mechanisms reduce the field error — the decomposition notably, from 0.183 to 0.110 — while leaving the wake collapsed. What restores shedding is enforcing causality: solving a short window, then using its end state as the initial condition for the next, which strengthens the initial condition relative to the bulk residual and forbids information from flowing backwards in time.

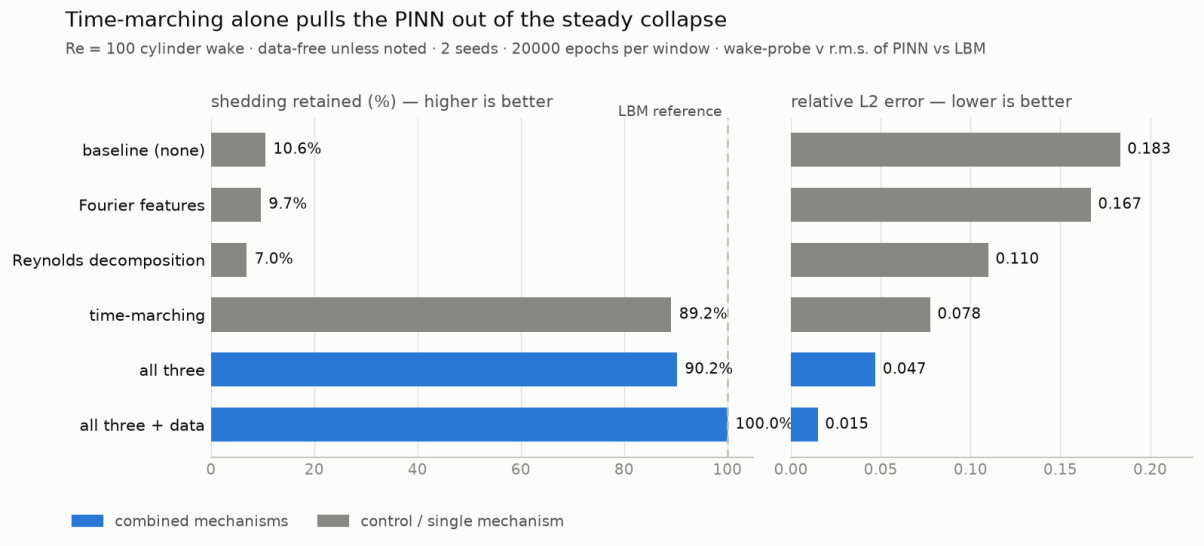


Figure 3: The ablation. Neither Fourier features nor the Reynolds decomposition escapes the collapse; causal time-marching does.

5.2 The equal-budget control

A marched run of four windows takes four times the optimiser steps of a single-window run, so the comparison above is confounded by compute. Re-running the single-window configurations at 80 000 epochs — the same total — settles it:

Table 6: Equal-budget control. Both improve; neither approaches 89%. *The gain is causality, not compute.*

configuration	20 000 ep, 1 window	80 000 ep, 1 window	$4 \times 20\,000$ ep, 4 windows
`baseline`	10.6%	21.9%	—
`decompose`	7.0%	23.8%	—
`march`	—	—	**89.2%**

This control is essential to the claim and should be quoted with it; without it, "time-marching wins" is indistinguishable from "time-marching trains longer".

5.3 Hyperparameters, and one that mattered only in combination

Table 7: Knobs, two seeds each.

change	applied to	shedding retained	relative L_2
$\sigma = 8$ (from 2)	`all`	90.2% → **97.6%**	0.047 → **0.042**
$\sigma = 8$	`fourier`	9.7% → 11.9%	0.167 → 0.147
$w_{ic} = 100$ (from 10)	`all`	90.2% → 89.8%	no change

The Fourier bandwidth σ is worth several points *on top of* the mechanism that works, and nothing at all without it. The initial-condition weight, by contrast, changes nothing once marching supplies the causal chain — consistent with the interpretation that marching works by strengthening the initial condition, making further weighting redundant.

5.4 How much supervision is actually needed

Table 8: Supervised points per window, two seeds, seed means.

data points per window	shedding retained	relative L_2
0	90.2%	0.047
100	96.2%	0.030
500	97.6%	0.021
2000	99.5%	0.016
4000	100.0%	0.015

Shedding saturates by 100 points per window; everything beyond that buys accuracy only. *The physics buys the dynamics; the data buys the last factor of three in error.*

6 How accurately does it shed?

Shedding retained is an amplitude ratio: it answers "is there a wake", which is the right question for the collapse and the wrong one for prediction accuracy. A network could hold the right amplitude at the wrong frequency, or drift out of phase within the window, and the ablation table would not notice. The best data-free configuration was therefore compared against the reference probe signal directly.

Table 9: Prediction accuracy of the best data-free configuration (`all`, $\sigma = 8$) at Re = 100, over one shedding period and over five.

quantity	1 period (4 windows)	5 periods (20 windows)	reference
shedding frequency St	0.1565 (-0.9%)	**0.1622 ($+0.1\%$)**	0.1621 (LBM)
wake amplitude	96.5%	94.8%	LBM probe r.m.s.
phase lag	$+2.9°$	$+2.4°$	—
phase drift across span	-0.000 periods	-0.005 periods	—
velocity-field error	0.043	**0.036**	relative L_2
zero crossings available	2	10	12+ for the LBM's own St

Extending the run from four windows to twenty — from one shedding period to five — *improves* every quantity that matters. The frequency error falls to $+0.1\%$ on ten zero crossings rather than two, the field error drops from 0.043 to 0.036, and the phase drifts by 0.005 of a period (about $2°$) across the entire five-period span. The marching chain therefore does not accumulate error in any visible way over the horizon tested: the wake stays locked to the reference in both frequency and phase, and only the amplitude gives up a little (96.5% to 94.8%).

Caveat. Ten zero crossings is enough to state St to about a percent, but it is still a shorter record than the LBM's own validation ($\sim$12 periods of a signal that is not itself being predicted). The frequency agreement to $+0.1\%$ should be read as "within the resolution of this measurement", not as a claim of three-digit accuracy.

Visible artefact. The probe trace shows small discontinuities at window handoffs, where each window starts from the previous one's end state rather than continuing it smoothly. This is the cost of the mechanism that makes the wake survive, and is reported rather than smoothed away.

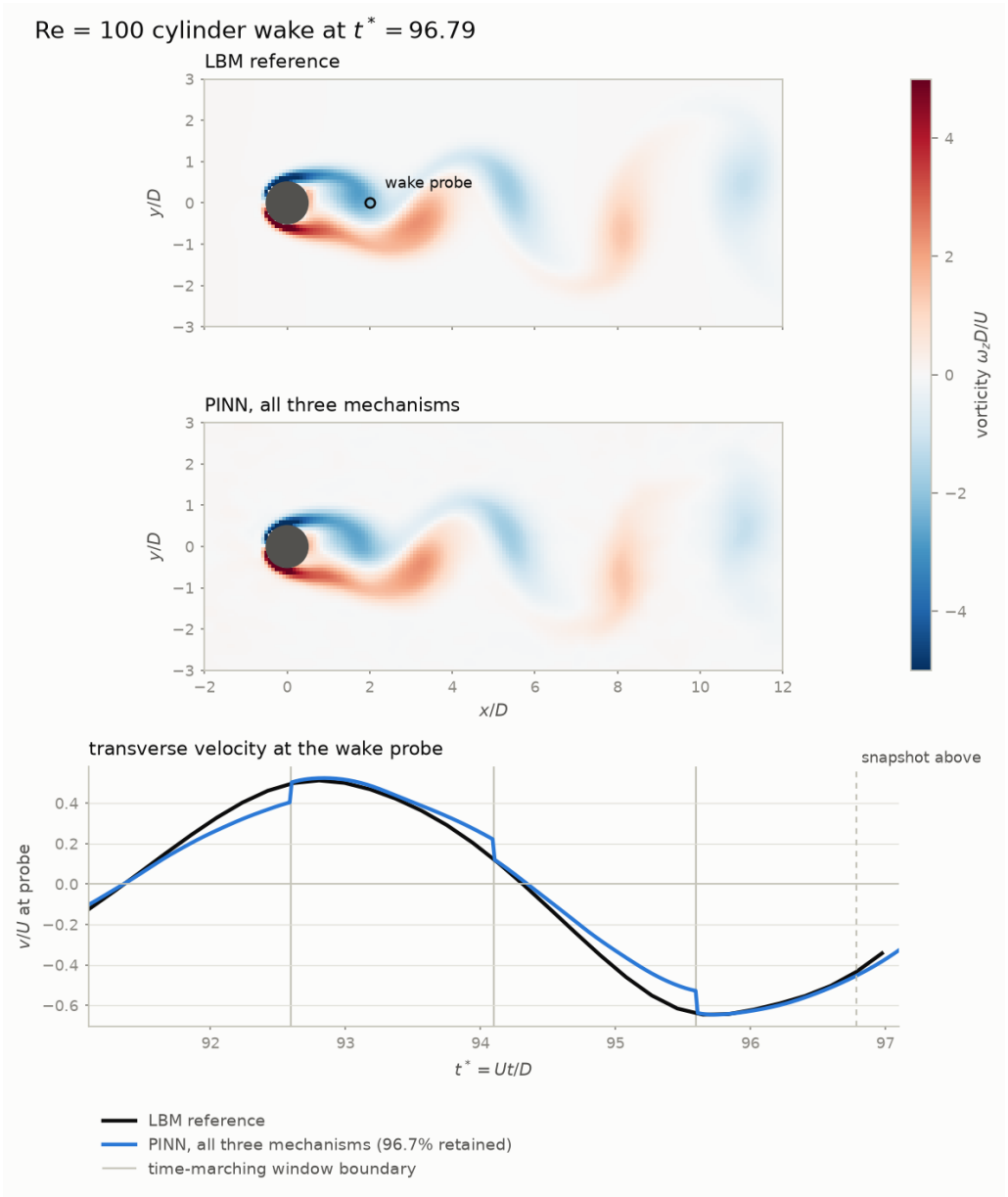


Figure 4: The data-free PINN against the reference. A Kármán street from a network that never saw data. The small steps in the probe trace are the time-marching window handoffs.

7 Does it hold across Reynolds number?

Everything above is at Re = 100. The comparison was therefore repeated at five more Reynolds numbers spanning the two-dimensional shedding range, with fresh ground truth generated and validated at each. The marching window is set to a quarter of each dataset's *measured* shedding period, so all configurations span exactly one period whatever the frequency — otherwise the comparison would not be like-for-like across Re.

Table 10: The ablation repeated across the shedding range. Two seeds per entry; Re = 100 is the main ablation, same configuration.

Re	St vs correlation	`baseline`	`all` (data-free)	relative L_2
60	+0.6%	37.8, 40.1%	**88.5, 91.2%**	0.031
80	−1.1%	22.2, 21.4%	**95.8, 98.5%**	0.022
100	−3.0%	9.8, 11.3%	**96.7, 98.4%**	0.042
120	+2.5%	20.2, 26.5%	**103.5, 104.1%**	0.035
150	−0.3%	35.7, 43.1%	**94.1, 96.7%**	0.032
180	−1.6%	31.1, 38.3%	**93.6, 96.3%**	0.027

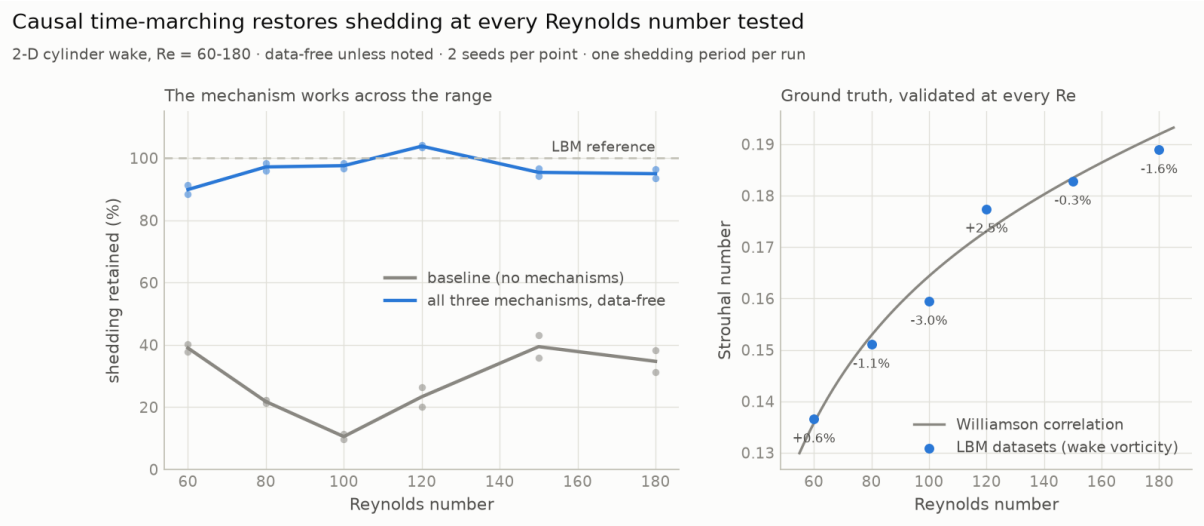


Figure 5: Left: shedding retained against Reynolds number, every seed plotted. Right: each dataset's own validation, measured Strouhal number against Williamson's correlation.

The result generalises. The control keeps 10–43% of the wake and the mechanism-equipped configuration 88–104%, at every Reynolds number tested, with field errors of 2–4%. The claim is therefore about the shedding regime rather than about one dataset.

Three things not to over-read. Re = 120 exceeds 100%: the network's wake is slightly *stronger* than the reference, which is amplitude overshoot, not extra accuracy. The `baseline` column varies from 10% to 43% with no clean trend and a seed spread of 5–8 points, so its variation should not be given a physical reading. And the Re = 60 and 80 datasets run at a lower Mach number than the rest, for the reason in Section 7.1 — closer to incompressible, but not an identical setup.

7.1 An acoustic resonance that captured the low-Re wakes

The first attempt at this sweep produced Strouhal numbers that fell as Reynolds number rose — 0.114 at Re = 60, 0.113 at Re = 80, 0.159 at Re = 100 — which is impossible for this flow, and worth recording as a methodological result.

The cause is not the measurement. Wake *vorticity*, which cannot see irrotational sound, oscillates at the same anomalous frequency. Lattice Boltzmann is weakly compressible and the free-slip walls reflect perfectly, so the domain supports a standing half-wave across its height at $c_s/2H$. In flow units that is St = 0.1155, and the distance from the shedding frequency decides whether the wake locks onto it:

Re	separation from the resonance	outcome
60	15%	locked, wake at 0.114
80	24%	locked, wake at 0.113
100	30%	clean, −3.0% against the correlation
120–180	33–40%	clean, −1.6% to +2.5%

The resonance sits at $(c_s/2H)(D/U)$ in flow units, so *lowering* the lattice velocity moves it away from the shedding band rather than merely weakening it. Regenerating Re = 60 and 80 at $u = 0.0315$ and 0.028 (against 0.05 elsewhere) moved them to +0.6% and −1.1% against the correlation. The same resonance explains a long-standing oddity in this project: the lift coefficient's r.m.s. never converged between runs, because its amplitude is modulated by the acoustic mode beating against the shedding at *every* Reynolds number, including Re = 100.

Consequently all Strouhal numbers reported here are measured from wake vorticity averaged over many points, with a single-point probe velocity as an independent check — the two agree to about 1% at every Re. The lift signal is retained for C_d, which it measures well, but is no longer trusted for frequency: at Re = 80 it reads 0.1619 where the vorticity says 0.1511 and the correlation 0.1528.

8 Physics in a neural operator

The reference paper's FNO is purely data-driven. This section reproduces its recipe and then asks whether physics improves it.

Reproduction. `underPINN`'s `FNO2D`, trained after Renn et al.: Reynolds-decompose first, map two consecutive frames to the next, apply recursively for ten steps during training with the loss summed over every step; width 80, 24 modes, Adam at 1×10^{-3}, 200 epochs, batch 16.

Contribution. A physics residual is added to that loss. The network predicts (u, v) only, so any residual containing ∇p is unusable; taking the curl of the momentum equations removes the pressure and leaves vorticity transport,

$$\omega_t + u\,\omega_x + v\,\omega_y - \tfrac{1}{\mathrm{Re}}\nabla^2\omega = 0, \qquad \omega = v_x - u_y, \tag{4}$$

evaluated by finite differences on the grid. Continuity is available separately as the cheaper variant. Both residuals are normalised by the residual the LBM data *itself* leaves — the data is weakly compressible and second-order accurate, so driving a prediction below that floor would be fitting discretisation error, not physics.

Result. Physics lowers forecast error by 20% at the trained horizon and 28% at twice it, then *crosses over* at step 27–32 across seeds and variants, which is 0.8–1.0 shedding periods. Continuity alone buys nearly all of the benefit, so the effect is not specific to the vorticity term — useful, since continuity is the cheaper residual by far.

Table 11: Rollout error (relative L_2 on (u, v), held-out initial conditions, two seeds). The trained horizon is ten steps.

forecast steps in shedding periods	10 (trained) 0.31	20 0.61	30 0.92	50 1.53
`data` (paper recipe)	0.093	0.183	**0.296**	**0.365**
`physics` (continuity + vorticity)	**0.074**	**0.132**	0.307	0.772
`physics_div` (continuity only)	0.075	0.140	0.312	0.764

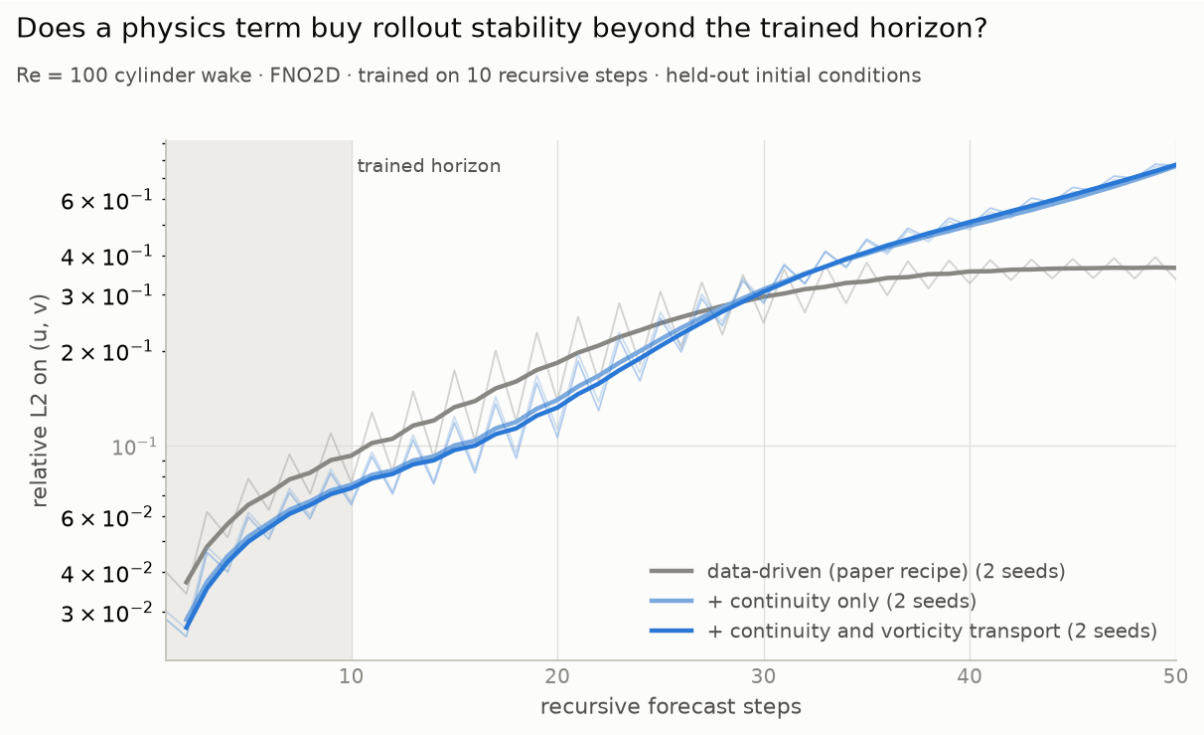


Figure 6: Forecast error against horizon. Bold lines are pairwise means over adjacent steps; faint lines are raw (see the parity note). The curves cross at approximately one shedding period.

Why the late-horizon ordering is not reported as a result. By 50 steps both variants have failed in the sense that matters: the error on the fluctuation exceeds unity, meaning that predicting zero fluctuation would beat either rollout. The data-driven curve flattens because it decays towards the mean flow — the safest wrong answer — while the physics-constrained rollouts keep injecting fluctuation energy and overshoot. Across seeds the physics variant reads 0.63 and 0.92 at 50 steps, so the gap is not reproducible. The defensible statement is that *physics extends the useful forecast horizon, and neither model is usable beyond one shedding period.*

A parity artefact worth naming. Rollout error alternates between even and odd steps by 30–45% in *every* variant, physics or not: a period-2 mode of the two-frame autoregressive map. A single step's error can therefore read 30% better or worse purely by which parity it lands on, so all numbers above are pairwise means of adjacent steps, and the figure keeps the raw curve visible.

Comparison with the reference paper. At the trained horizon the data-driven variant reaches $\epsilon = 0.093$ against the paper's $\epsilon < 0.1$. *This is not an improvement on their work.* Our data is clean, periodic, two-dimensional LBM output at $\mathrm{Re} = 100$; theirs is noisy experimental

PIV at Re = 240–3060 with genuine three-dimensionality. The only claim supported is that the reproduction works.

9 Conclusions

1. The documented collapse reproduces on validated ground truth: a data-free PINN retains 10.6% of the wake, and quadrupling the training budget does not fix it.
2. **Causal time-marching alone restores shedding, data-free** (89.2% retained), and an equal-budget control attributes the gain to causality rather than to compute. Fourier features and the Reynolds decomposition do not escape the collapse on their own, but improve accuracy substantially once marching is in place, taking the field error from 0.078 to 0.042.
3. The best data-free configuration predicts the shedding frequency to 0.9%, the amplitude to 96.5% and the phase to 3°, over one shedding period, with a velocity-field error of 4.3%.
4. Supervision is needed in far smaller quantity than expected: 100 points per window restores 96% of the shedding, and further data buys accuracy only.
5. Adding a pressure-free physics residual to a data-driven FNO lowers forecast error by 20–28% within one shedding period, with continuity alone responsible for nearly all of it. Beyond one period both variants fail.

10 Limitations and ongoing work

- **One Reynolds number.** Every result above is at Re = 100. A sweep over Re = $60, 80, 120, 150, 180$ is running, repeating the `baseline`-versus-`all` comparison at each, with the marching window scaled to each dataset's measured shedding period so the configurations stay comparable. The upper limit is set by the three-dimensional transition near Re $\approx$ 190.
- **Five shedding periods of PINN evaluation.** Section 6 covers five periods, over which the accuracy improves rather than decays. Behaviour over tens of periods, where a marching chain might accumulate phase error, is untested — and the cost grows linearly with the horizon, one window per quarter period.
- C_l **r.m.s. is not converged** across configurations (0.18–0.54 against a benchmark of $\sim$0.23) and is deliberately not quoted as a validation metric.
- **The resolution series was run at 5% blockage**, where blockage masks resolution. The blockage series is sound; a clean resolution answer would require repeating it at $H = 40D$.

A note on method. Estimates for each phase were recorded *before* running it. Three were wrong in ways that changed conclusions: the collapse was expected to need all three mechanisms (it needs one); the Fourier bandwidth was expected to be the quick win (it does nothing alone); and the FNO physics term was expected to help only beyond the trained horizon (it helps within it and hurts beyond). An earlier CPU sweep at a fifth of the budget supported the first of those wrong conclusions — *an under-trained ablation produces the wrong qualitative answer, not merely a noisier one.*

References

[1] P. J. Renn, C. Wang, S. Lale, Z. Li, A. Anandkumar and M. Gharib, *Forecasting subcritical cylinder wakes with Fourier Neural Operators*, arXiv:2301.08290, 2023.

[2] *Can physics-informed neural networks beat the finite element method?* — and related analyses of PINN failure modes in unsteady flow, arXiv:2306.00230, 2023.

[3] Aeroscience Computations and Analysis Lab, `underPINN`: a JAX framework for physics-informed neural networks.